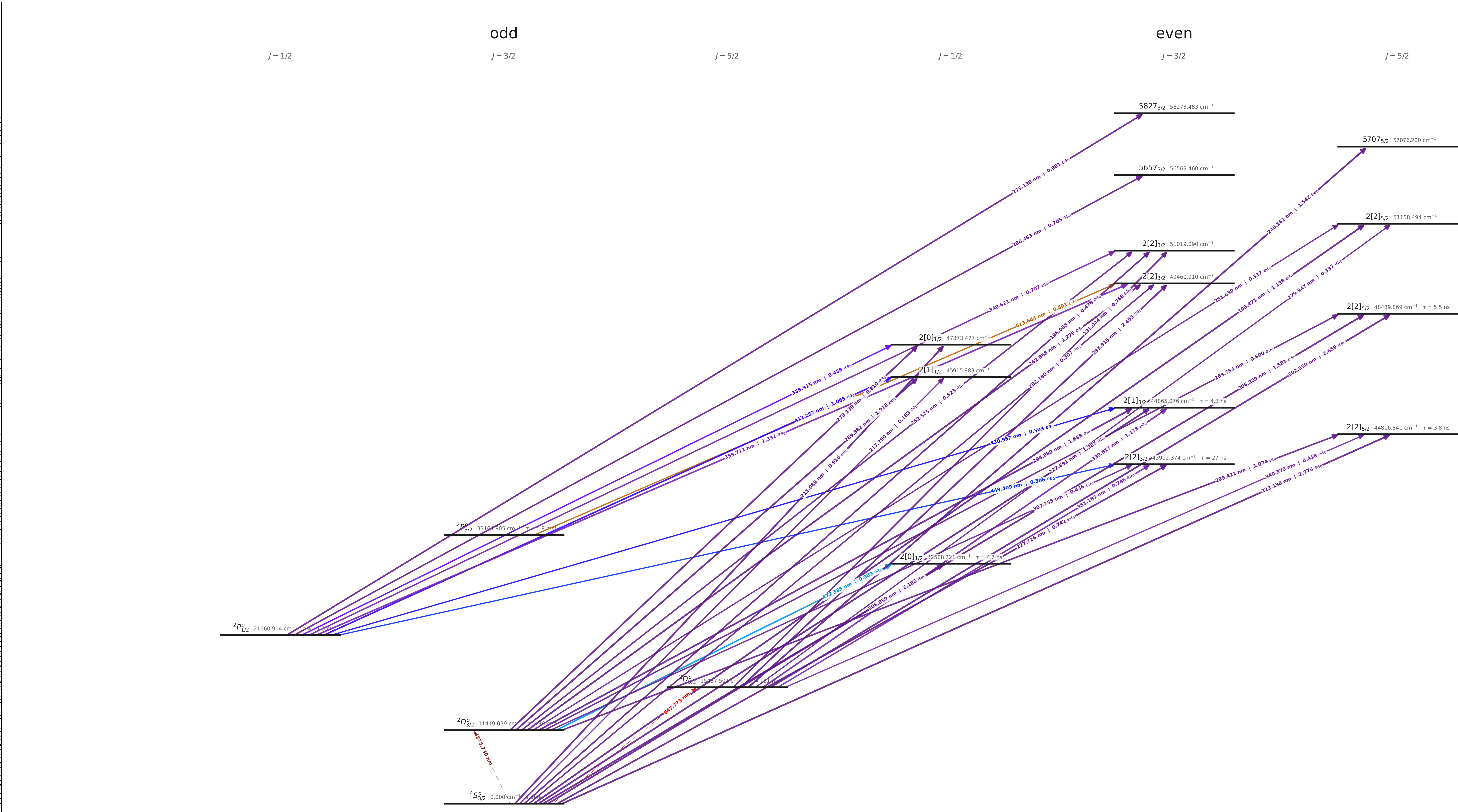


Bismuth

Bi (Bi I). Strongest classified transitions below 2 μm (NIST): 18 levels, 39 lines.

Bi⁺ ionisation limit 58761.65 cm⁻¹ = 7.28552 eV (170.179 nm)

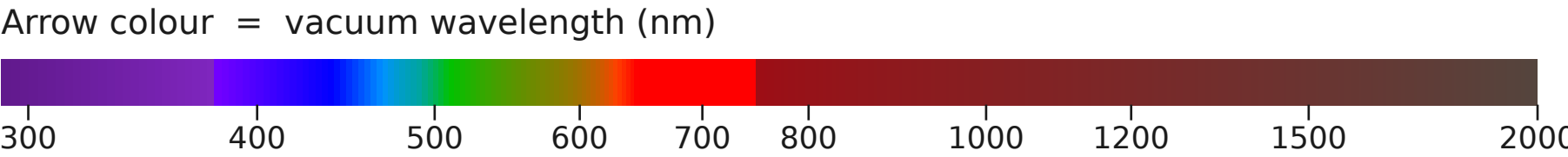


Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J e _r J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
⁴ S _{3/2} - 2[0] _{1/2}	306.8593	306.7702	976.9703	2.182	1.670e+08	33.86	78.5	resonance line for collinear and in-source laser spectrosc...
⁴ S _{3/2} - ² D _{5/2}	647.7732	647.5943	462.8046	-	-	1.054e-06	-	magnetic-dipole line used for parity-nonconservation opti...
⁴ S _{3/2} - ² D _{3/2}	875.7304	875.4900	342.3342	-	2.250e+01	4.421e-06	81	magnetic-dipole line of the Oxford parity-nonconservation...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||e_r||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



²⁰⁹Bi hyperfine structure (I = 9/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
⁴ S _{3/2}	-446.9420(10)	-304.6540(20)	F=3: +3547.64 F=4: +1962.97 F=5: -208.27 F=6: -3093.02
² D _{3/2}	-1231.020	-652.500	F=3: +9856.85 F=4: +5367.77 F=5: -651.39 F=6: -8472.51
² D _{5/2}	2502.860	23.000	F=2: -34403.78 F=3: -26902.10 F=4: -16897.18 F=5: -4386.71 F=6: +10632.17 F=7: +28162.92
² P ^o _{1/2}	11260.200	-	F=4: -30965.55 F=5: +25335.45
2[0] _{1/2}	4920.8(6)	-	F=4: -13532.20 F=5: +11071.80
² P ^o _{3/2}	491.028	978.640	F=3: -3602.44 F=4: -2290.75 F=5: -39.50 F=6: +3559.10

Measured A, B constants; sources in the data file.